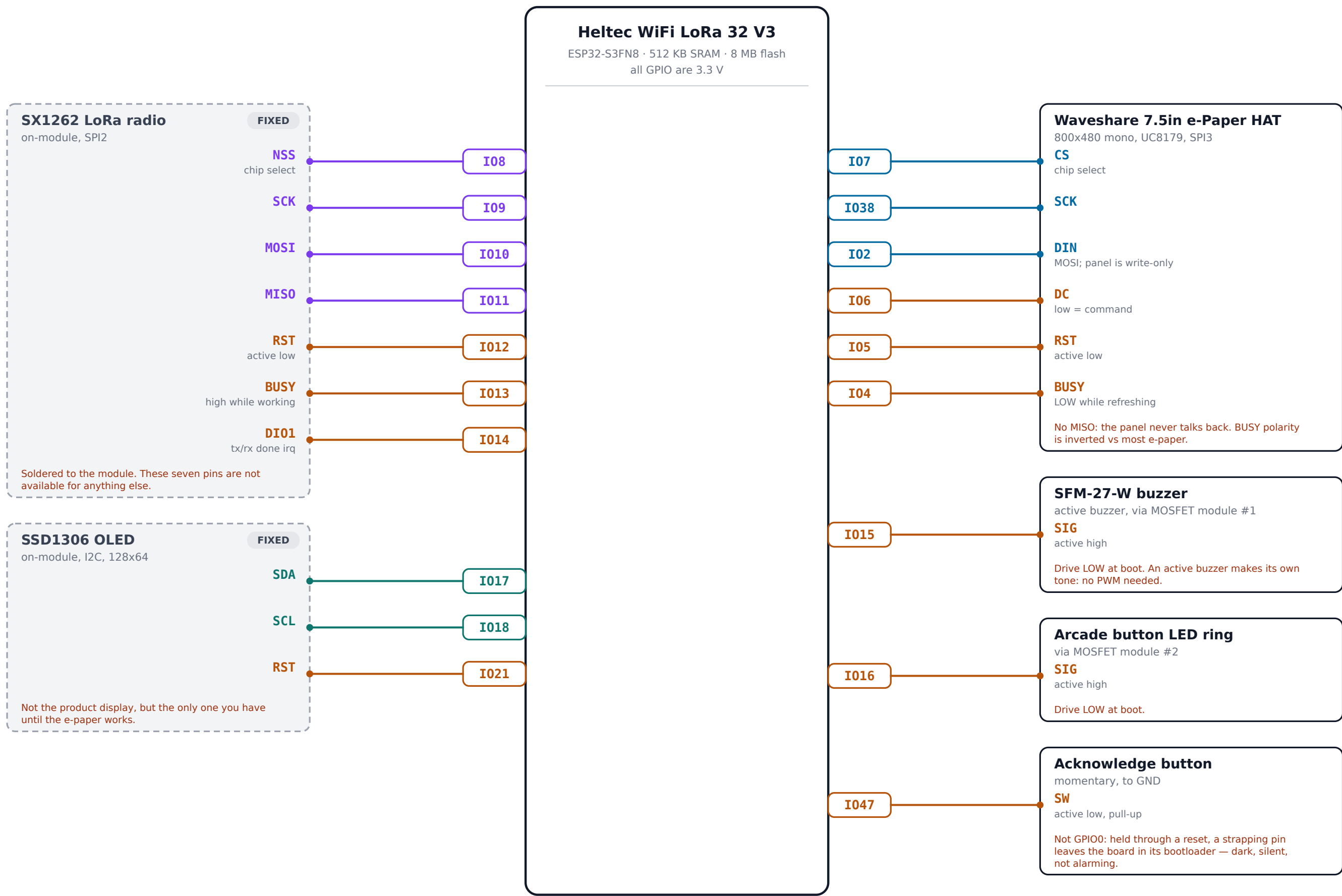


GlucoBeacon — Display Node

Heltec WiFi LoRa 32 V3 · ESP32-S3FN8 + SX1262 · Waveshare 7.5in e-Paper

wiring v1.2

```
generated from firmware/glucosebeacon-fw-display/src/board.rs
regenerate with tools/wiring/render.sh - do not edit by hand
```



Nets

- SPI2 — radio (on-module)
- SPI3 — e-paper
- I2C — OLED (on-module)
- control / GPIO

Radio link

region	US915 (902-928 MHz)
spreading factor	SF9
bandwidth	125 kHz
max frame	48 bytes

Both ends must match exactly. A mismatch is a silent link, not a weak one.

Pin map

SX1262 NSS	GPI08
SX1262 SCK	GPI09
SX1262 MOSI	GPI10
SX1262 MISO	GPI11
SX1262 RST	GPI12
SX1262 BUSY	GPI13
SX1262 DIO1	GPI14
OLED SDA	GPI17
OLED SCL	GPI18
OLED RST	GPI21
e-Paper CS	GPI07
e-Paper SCK	GPI038
e-Paper DIN	GPI02
e-Paper DC	GPI06
e-Paper RST	GPI05
e-Paper BUSY	GPI04
Buzzer SIG	GPI015
Button LED SIG	GPI016
Button SW	GPI047

Free / reserve

I00, I03, I033, I034, I039, I040, I041,
I042, I045, I046, I048

GPIO22-25 do not exist on an S3;
26-32 are the SiP flash.

Power

USB-C 5 V (2 A) and a 3.7 V LiPo meet at a power-path / OR-ing stage. The e-Paper HAT and both MOSFET modules run from 5 V; the Heltec regulates its own 3.3 V. All grounds common. Do not power the board without the antenna attached.

Changed from the hand-drawn v1.0

v1.1 — e-Paper SCK/MOSI/MISO were on GPIO12/11/13 and DC/CS on GPIO17/18: the first three are the SX1262, the last two the on-board OLED, so the bus moved to free pins. e-Paper MISO dropped, the panel is write-only. GPIO22/23 were listed as reserve; neither exists on an ESP32-S3. v1.2 — the acknowledge button moved off GPIO0 and the e-Paper clock off GPIO3. Both are strapping pins: held or pulled at reset they change how the chip boots, and a board stuck in its bootloader is dark, silent and not alarming.